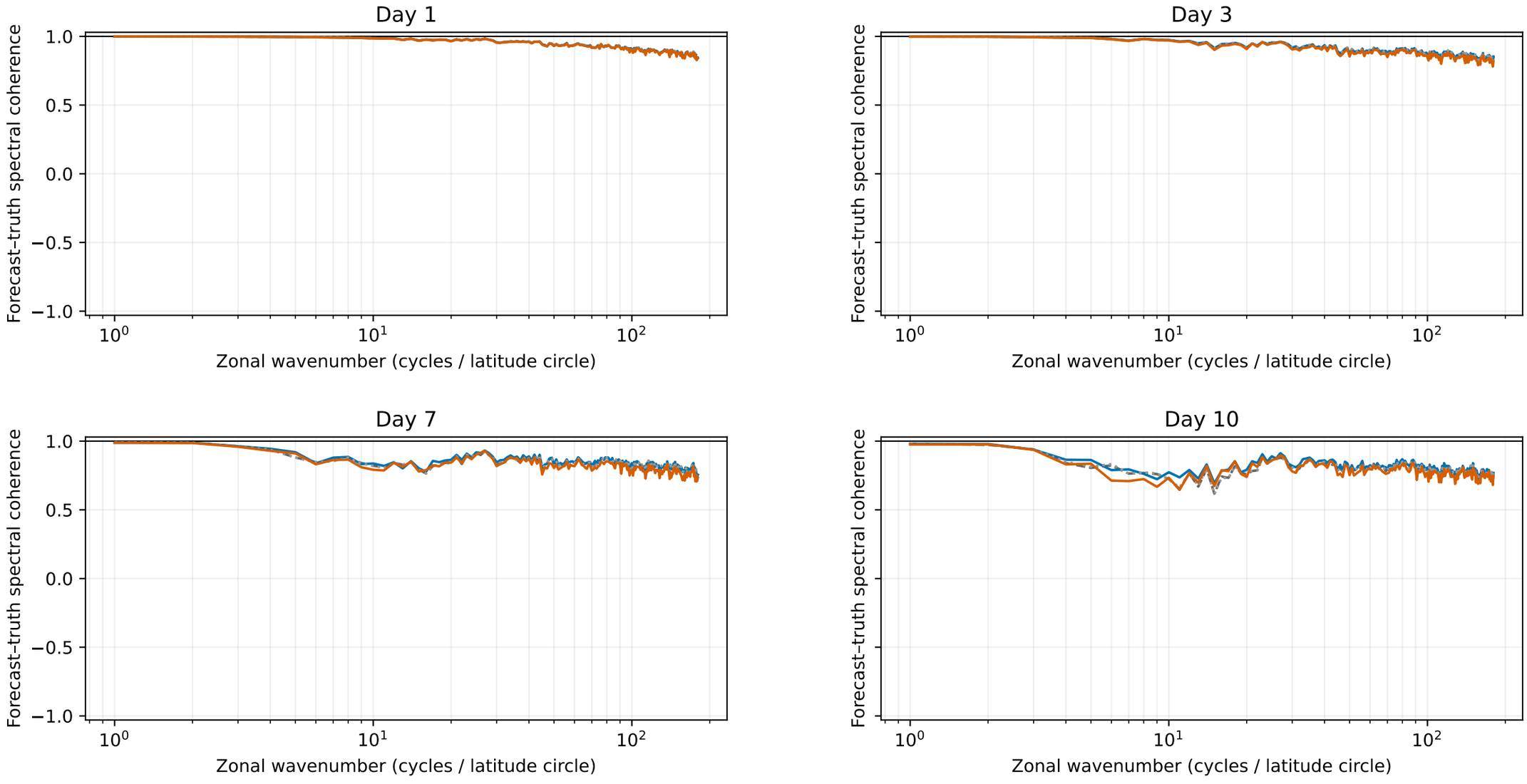
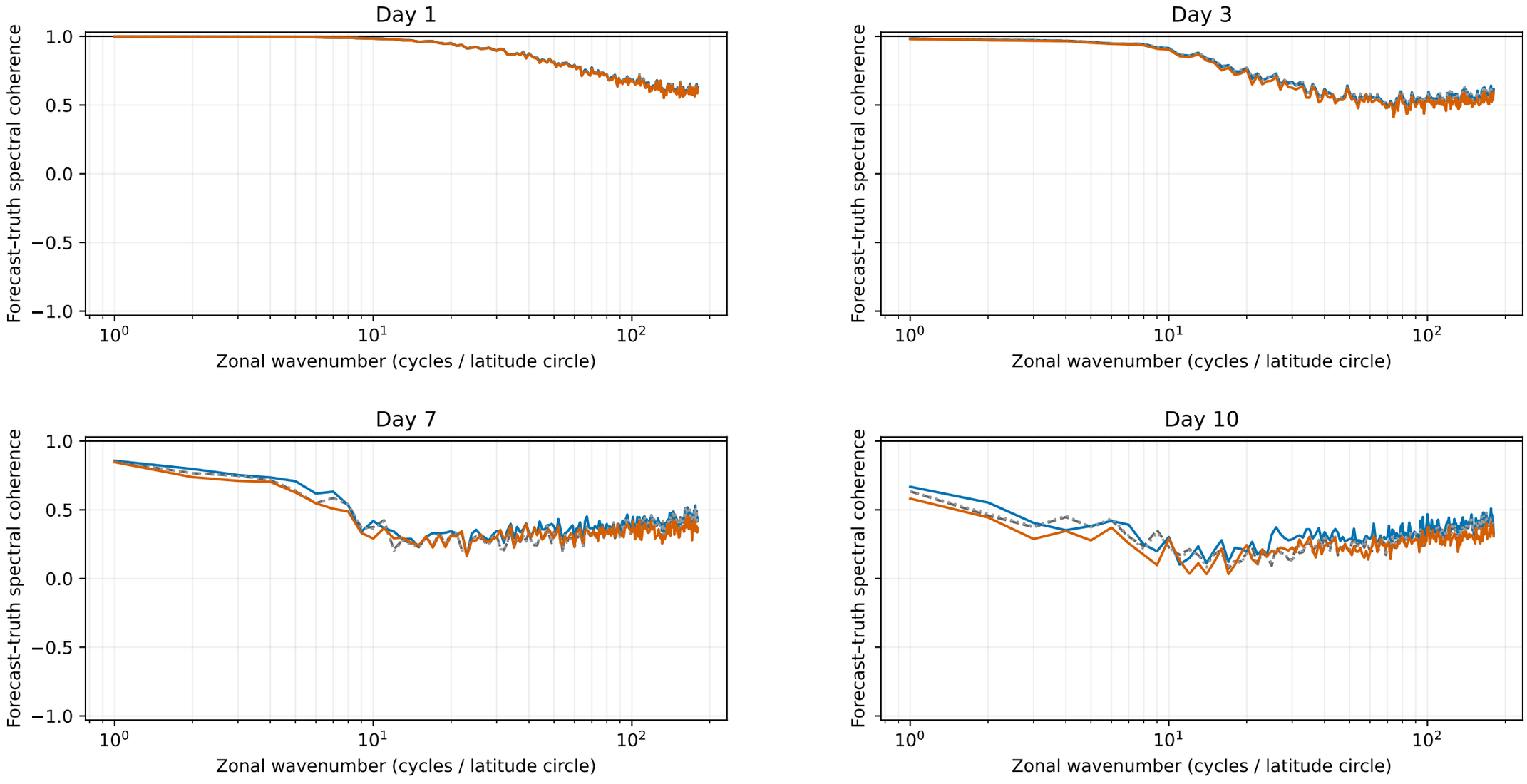
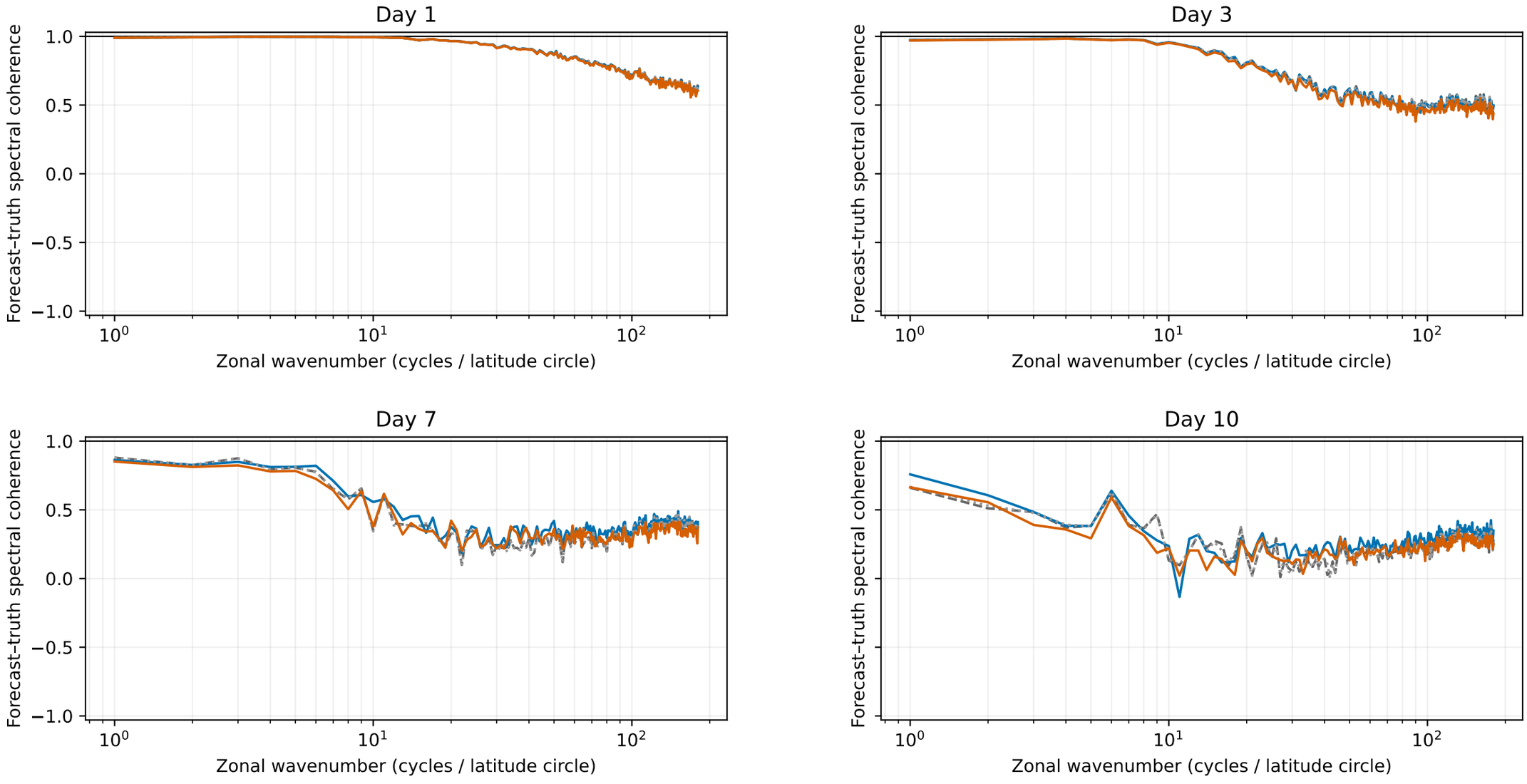
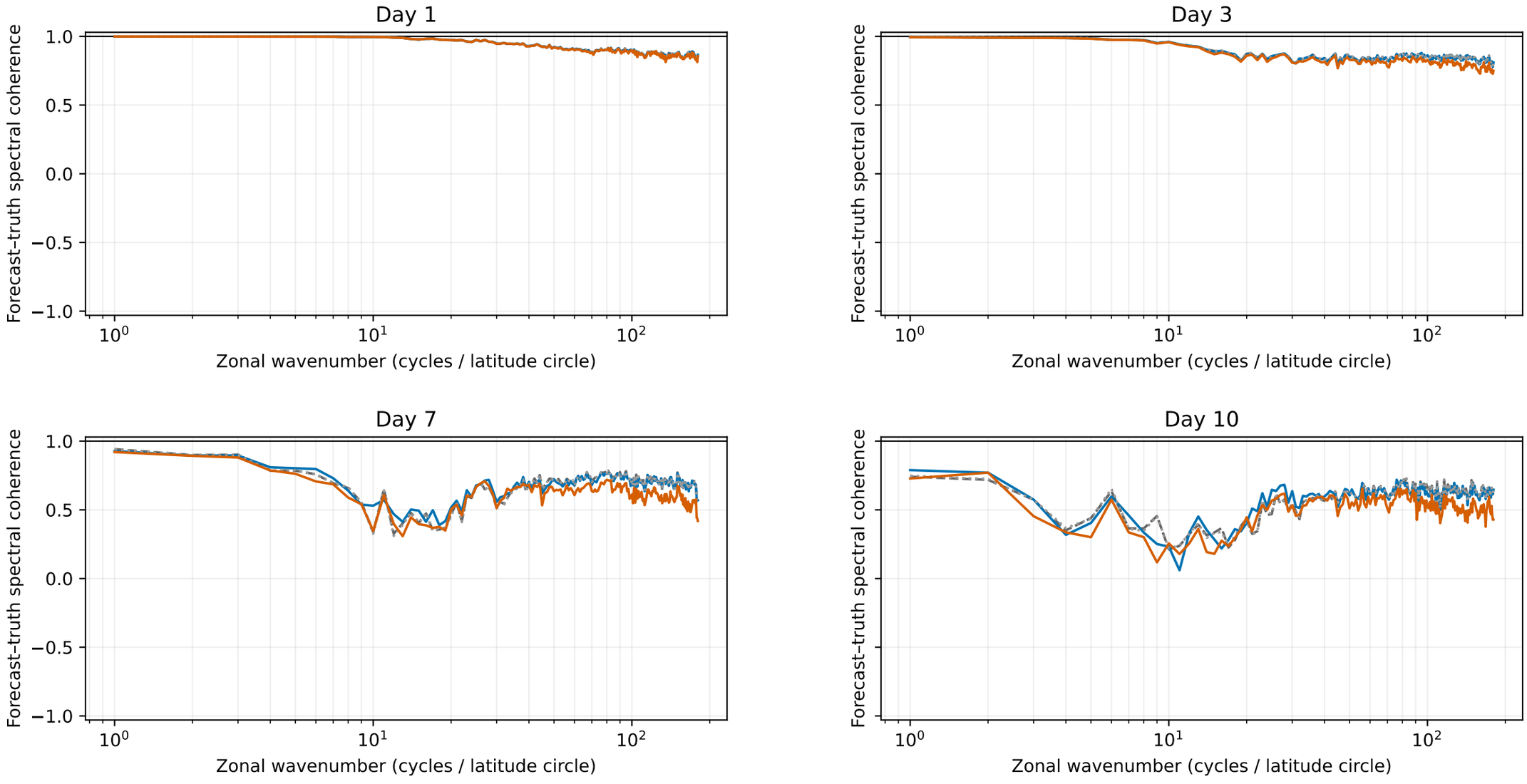
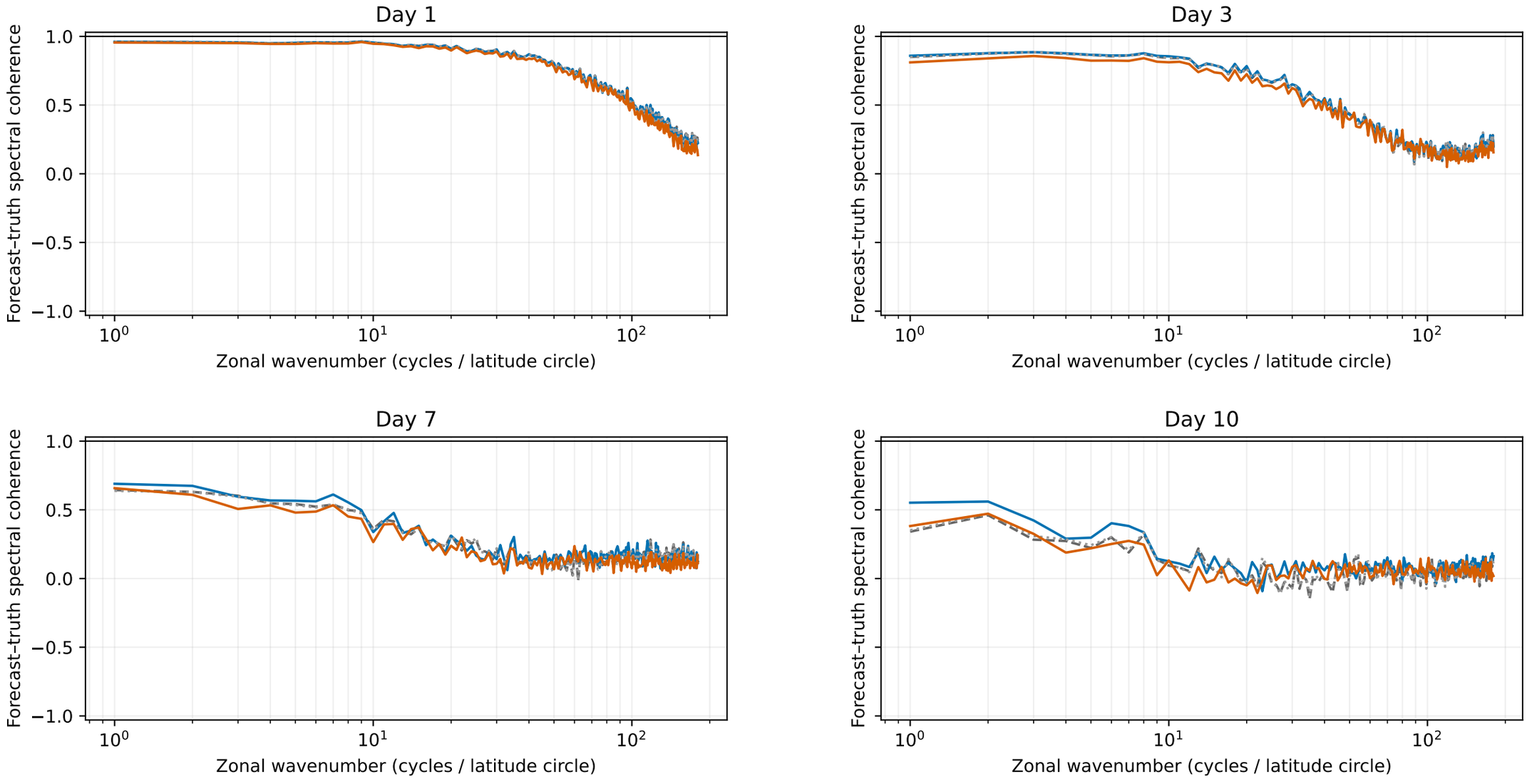
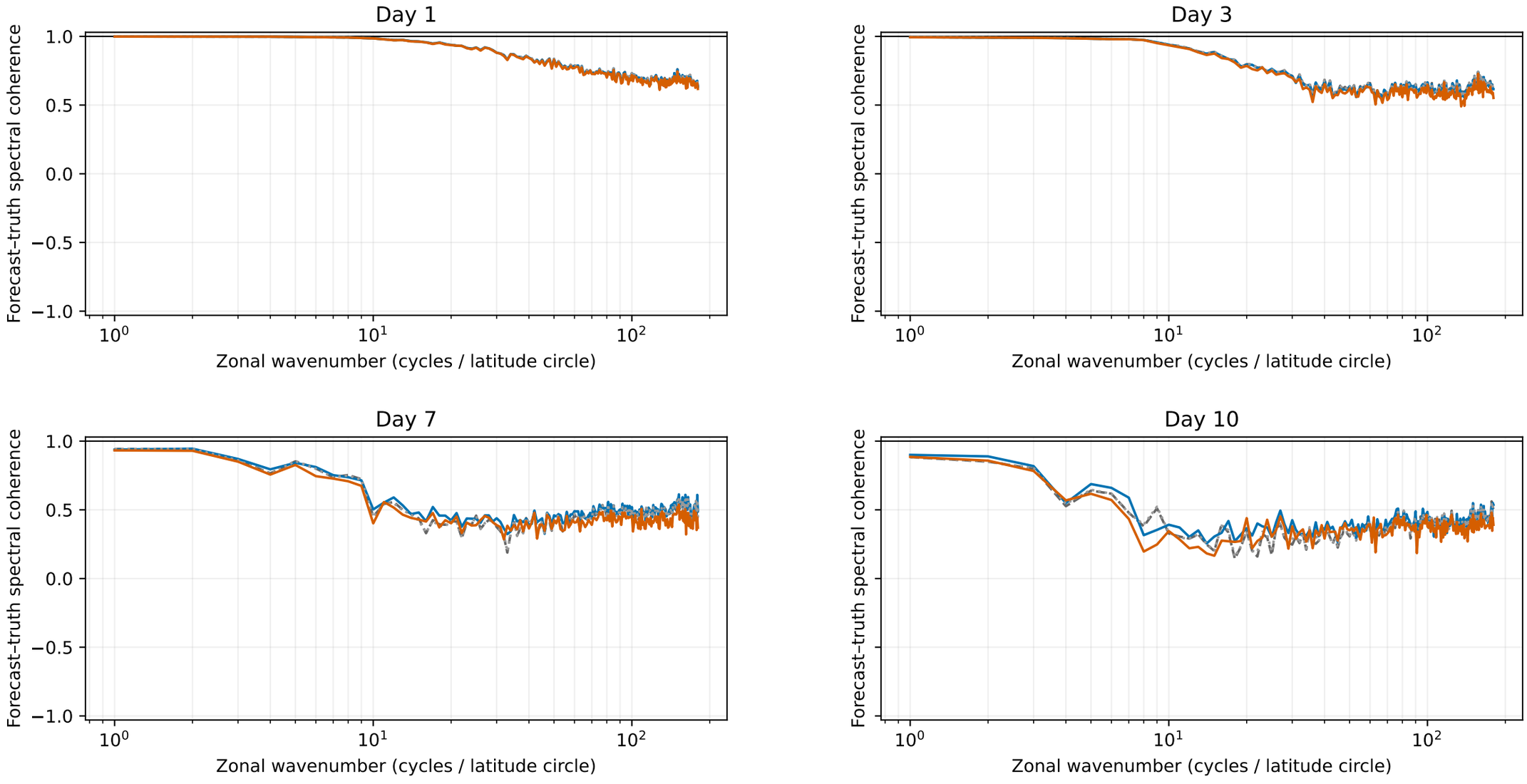

T2m: forecast–truth spectral coherence · step2000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

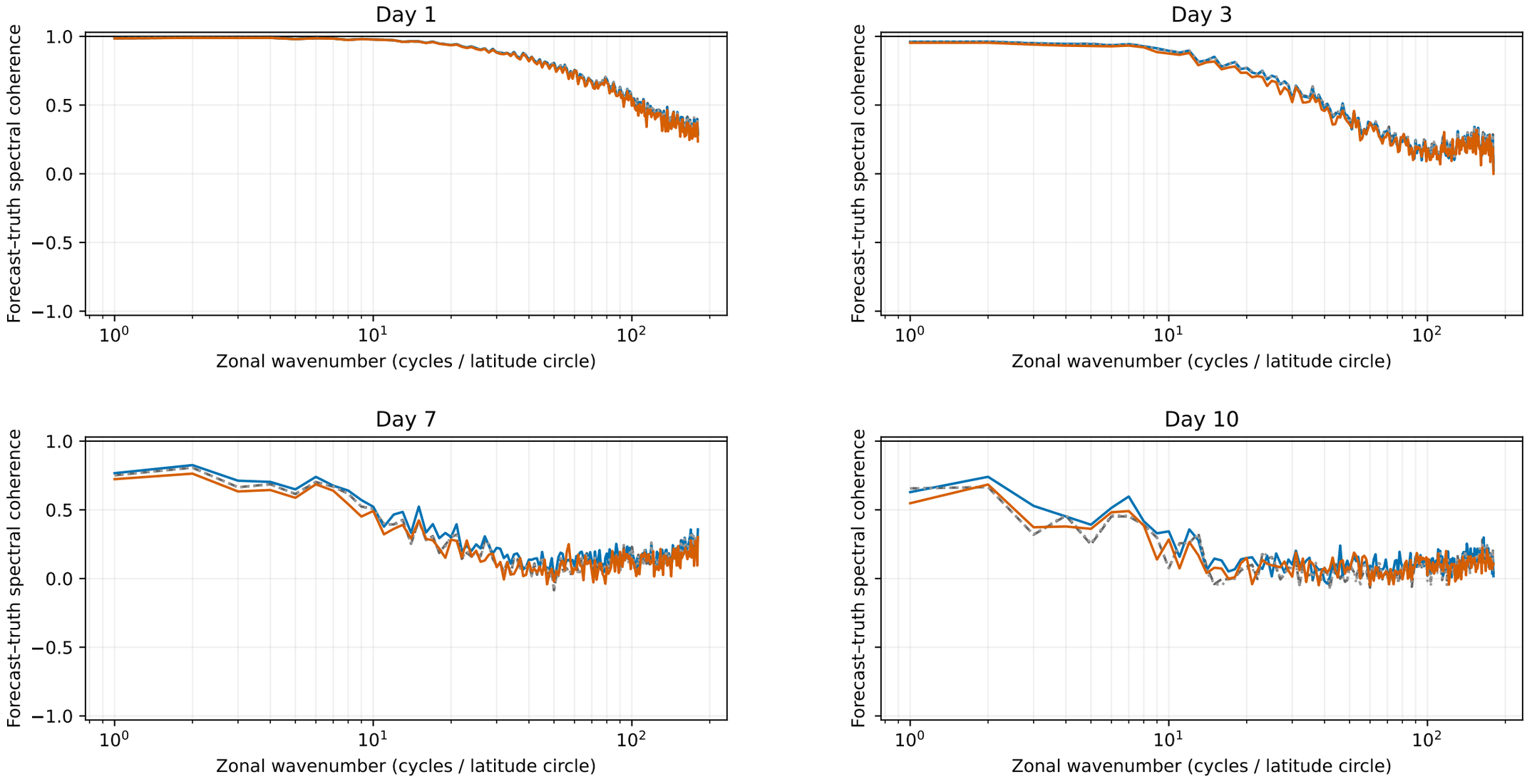

U10: forecast–truth spectral coherence · step2000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

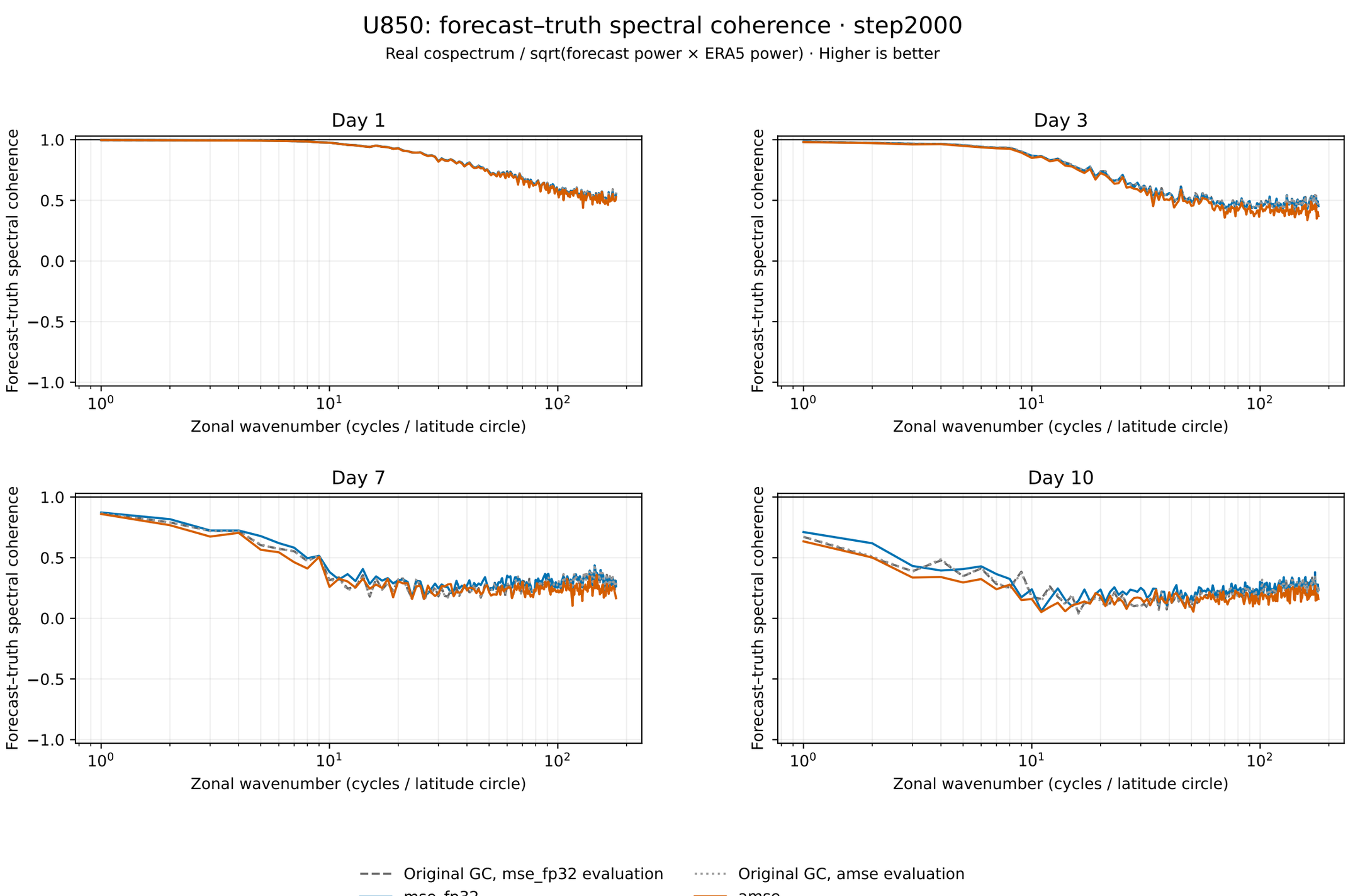

V10: forecast–truth spectral coherence · step2000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

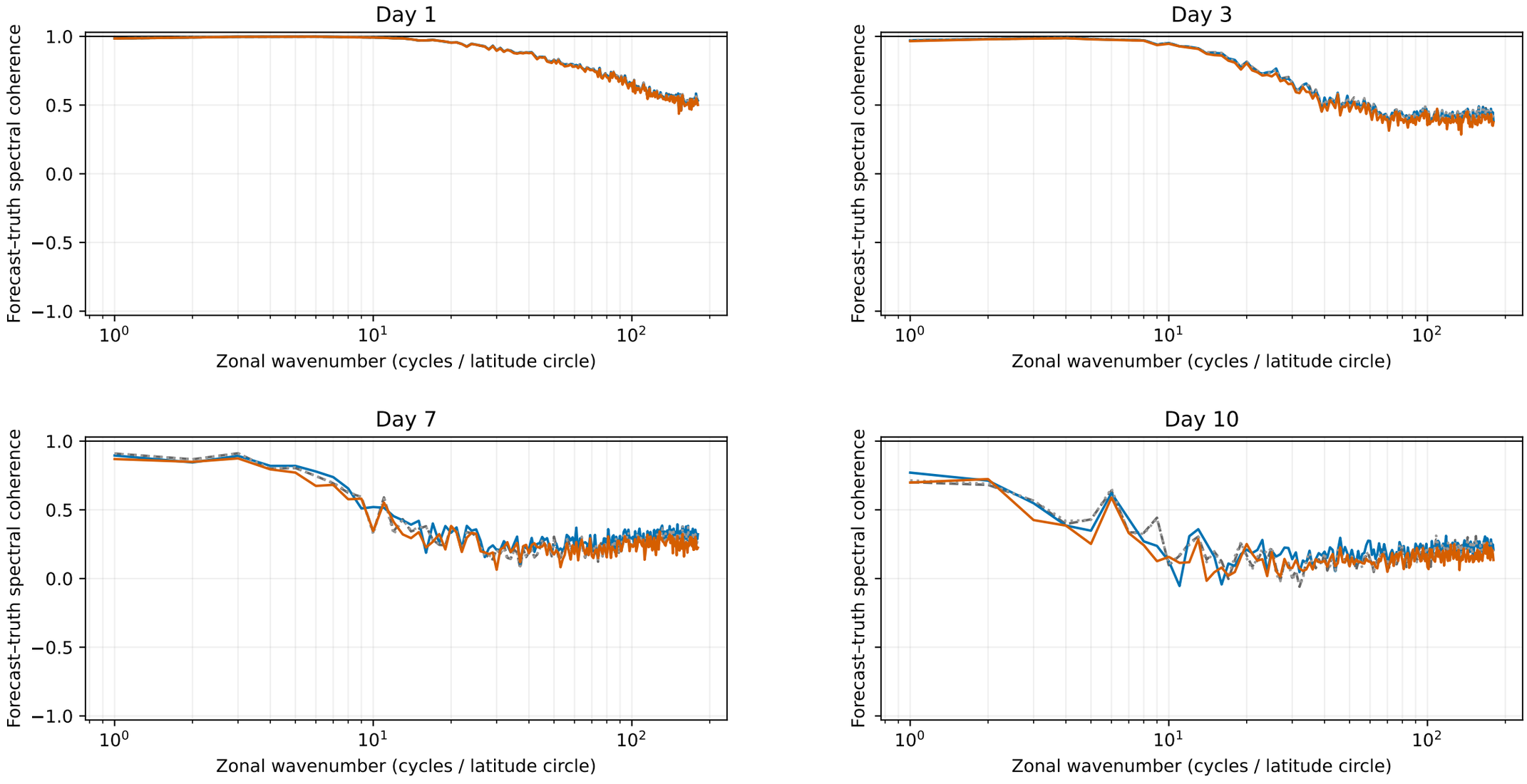

MSLP: forecast–truth spectral coherence · step2000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

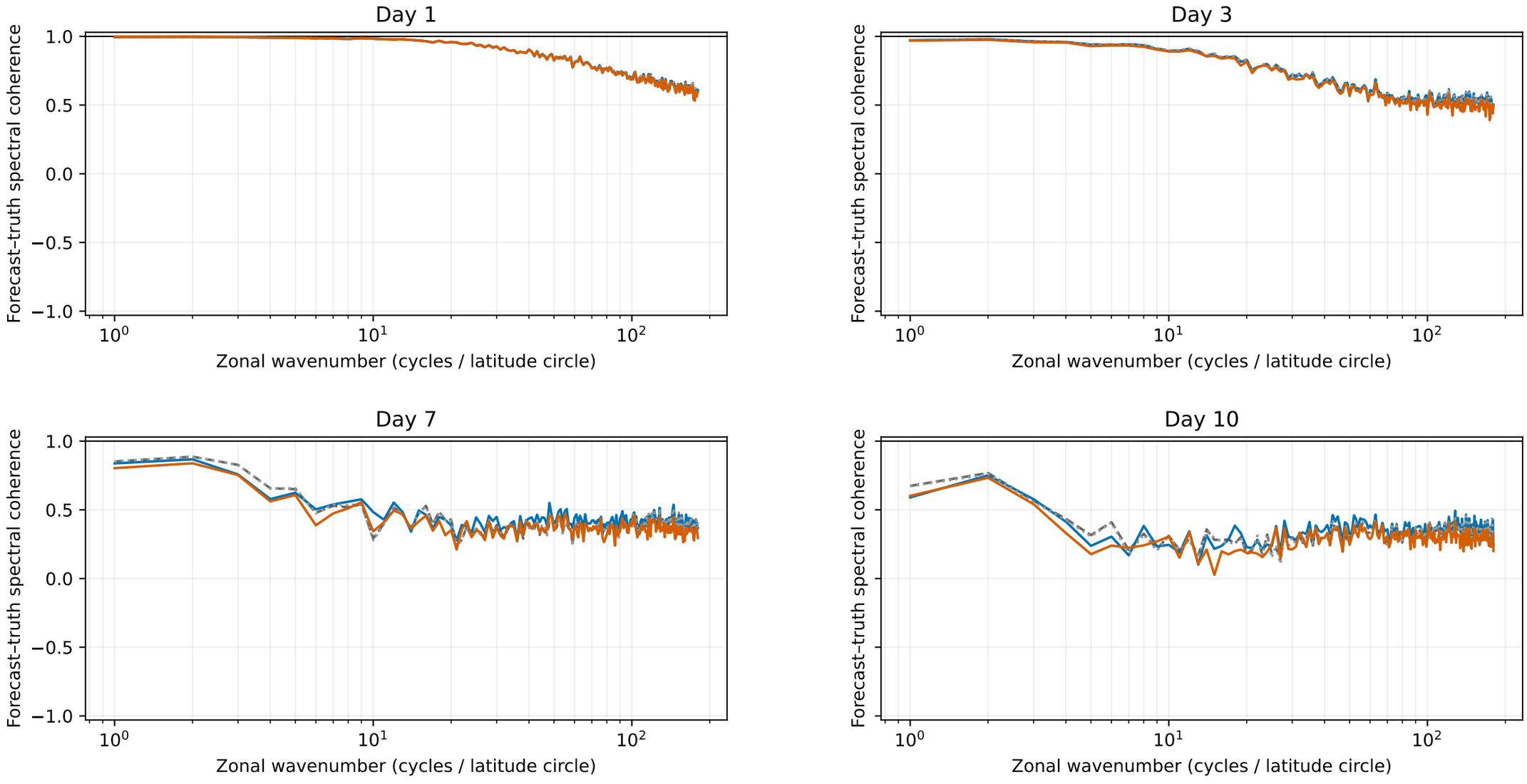

TP6: forecast–truth spectral coherence · step2000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
Zonal wavenumber (cycles / latitude circle)
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

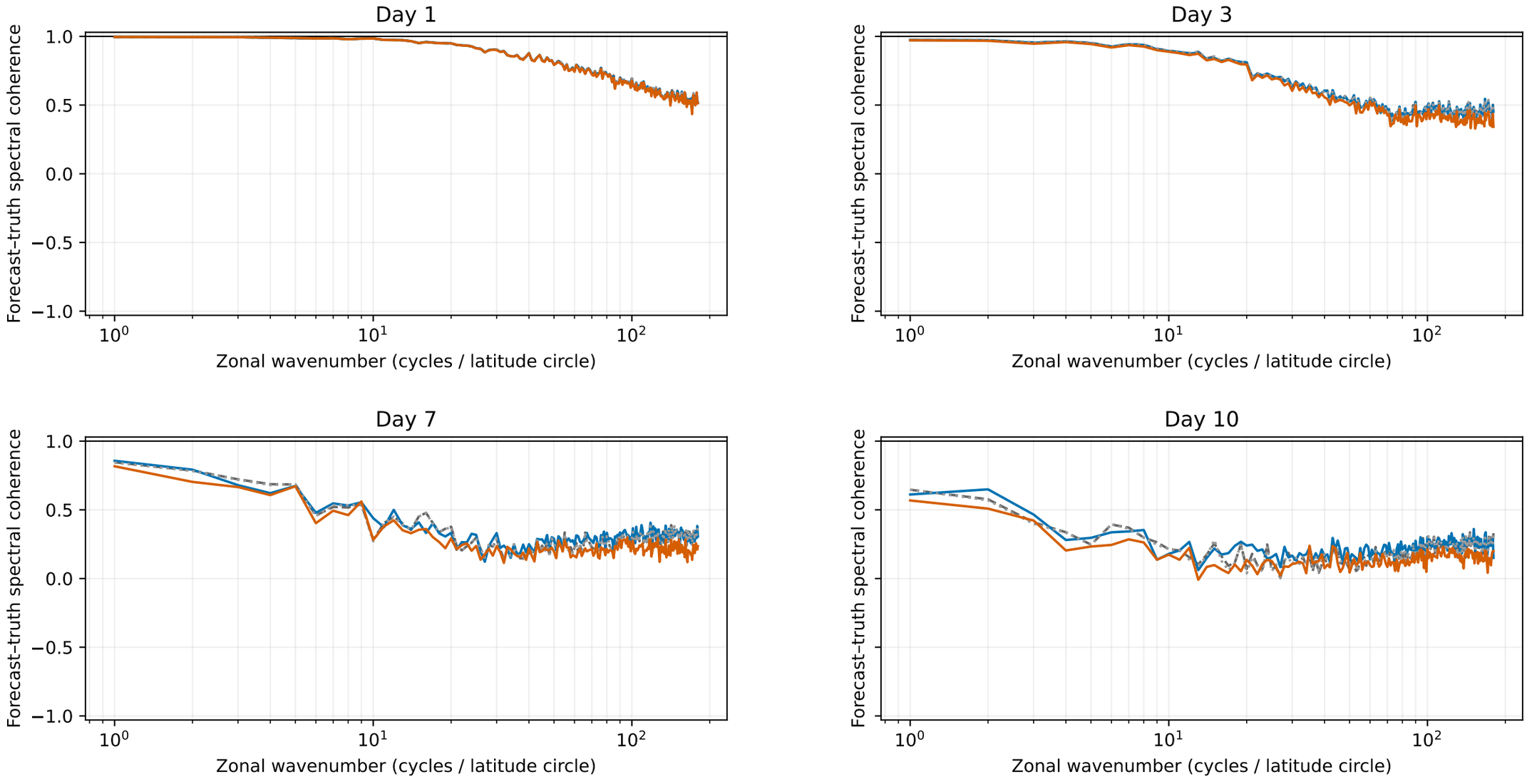

T850: forecast–truth spectral coherence · step2000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

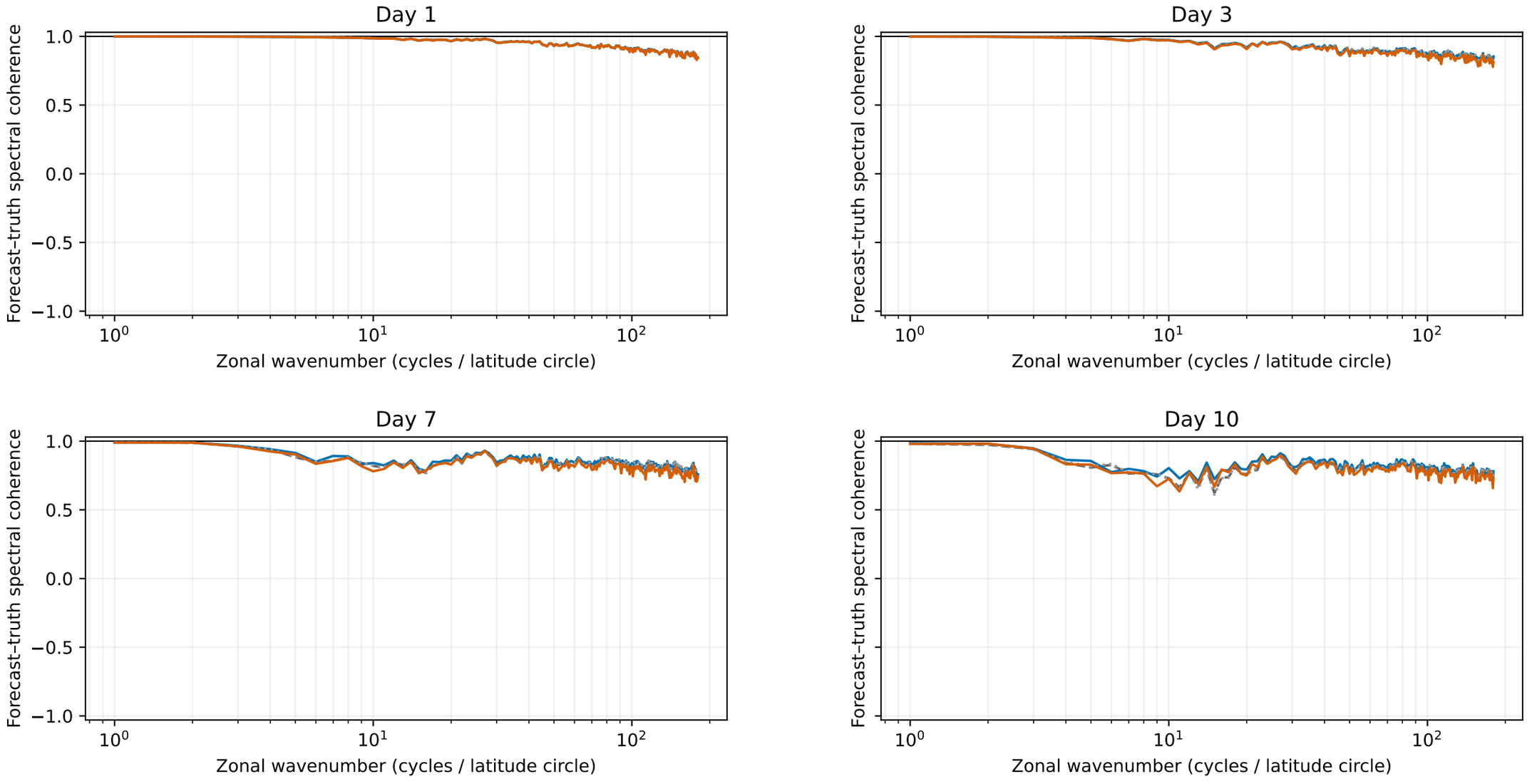

Q700: forecast–truth spectral coherence · step2000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
Zonal wavenumber (cycles / latitude circle)
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

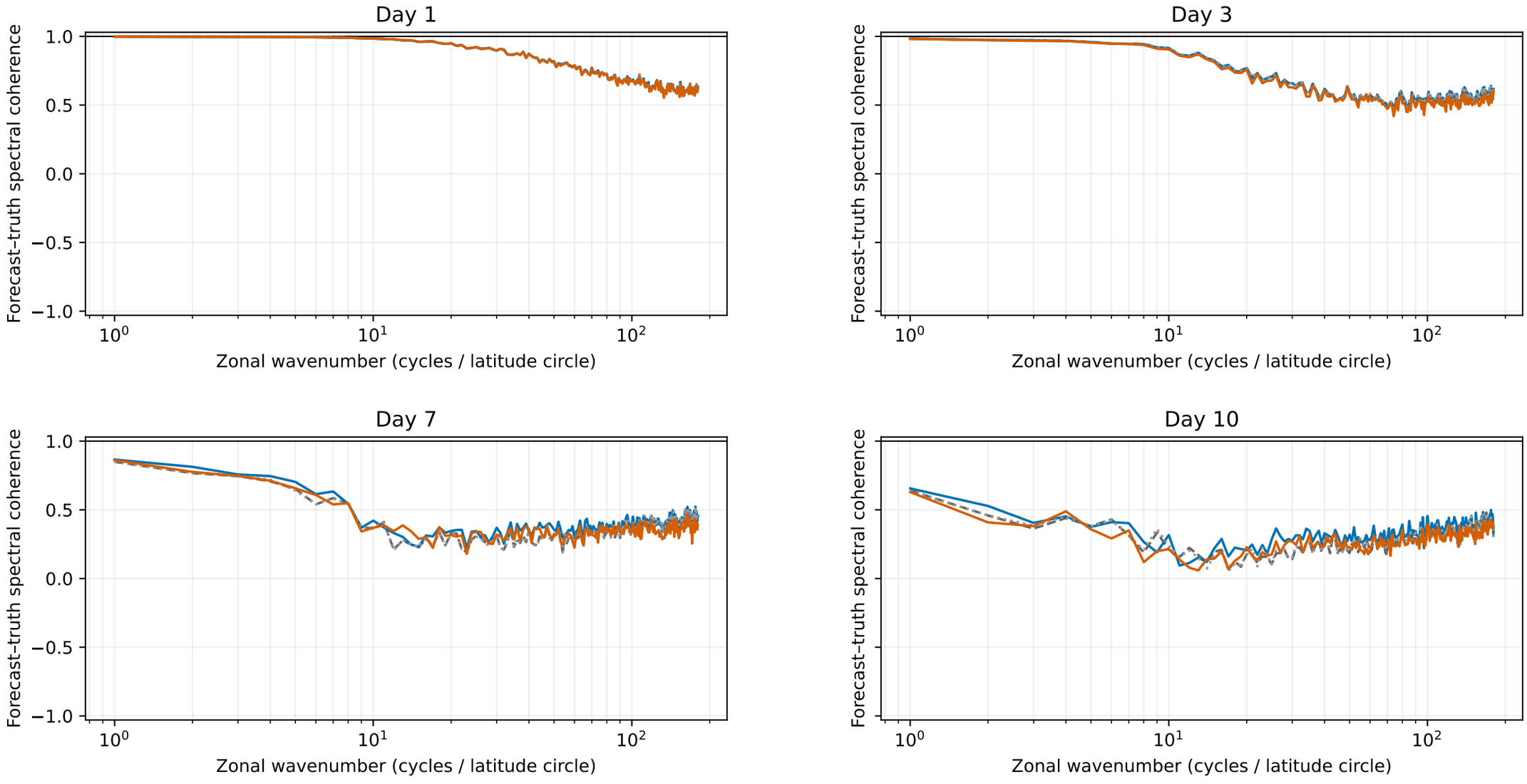

U850: forecast–truth spectral coherence · step2000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

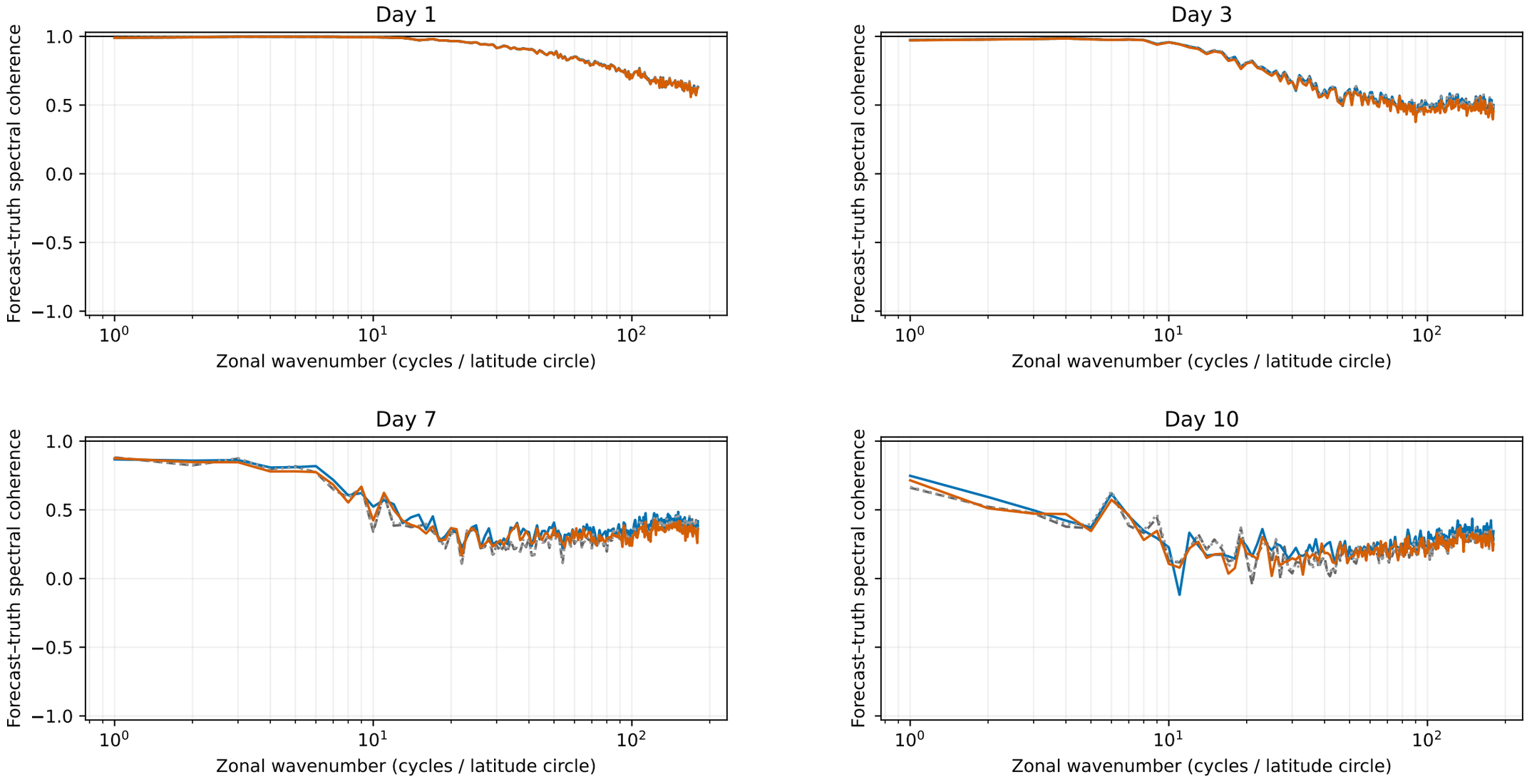

V850: forecast–truth spectral coherence · step2000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

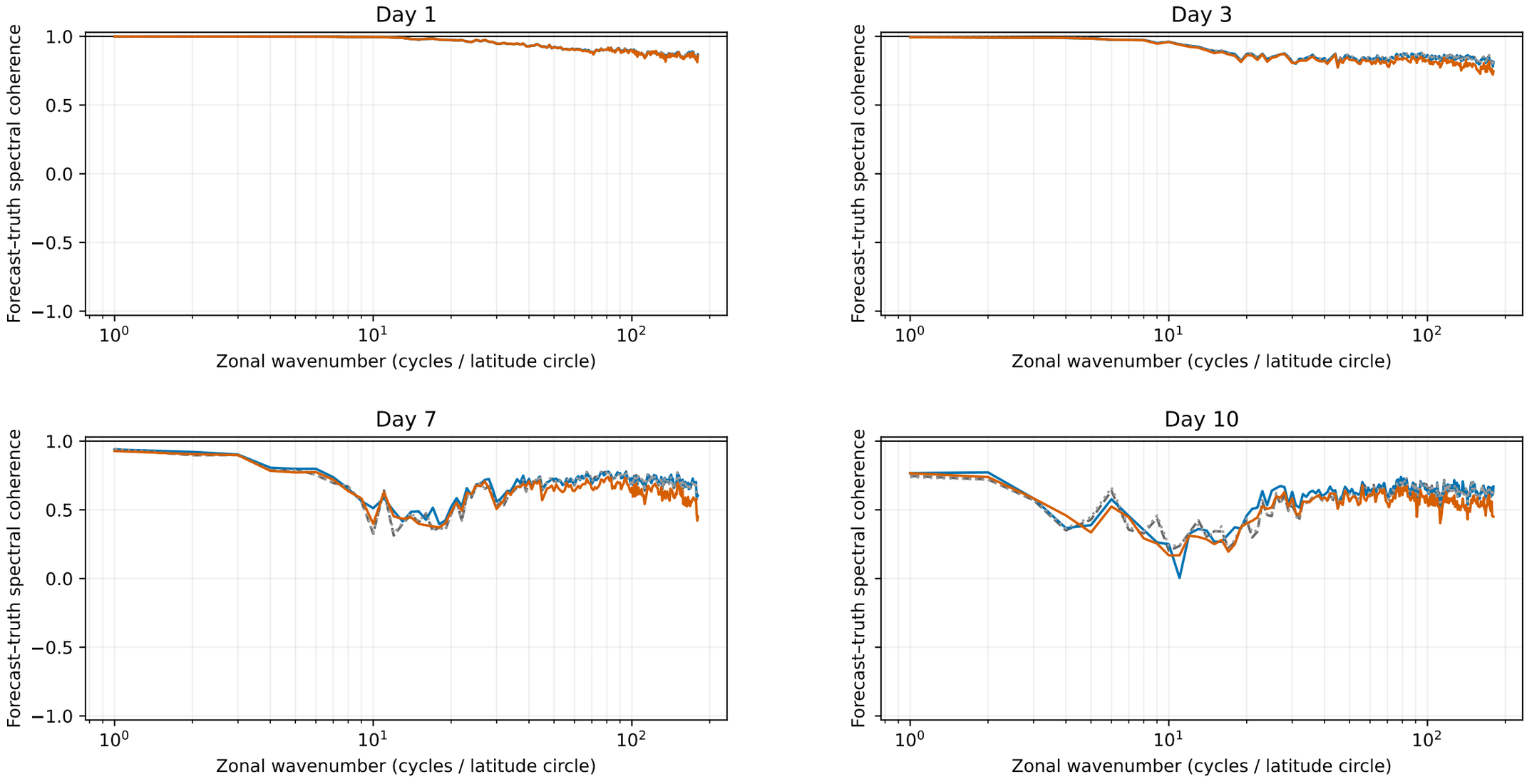

WS10: forecast–truth spectral coherence · step2000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
Original GC, amse evaluation
mse_fp32
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

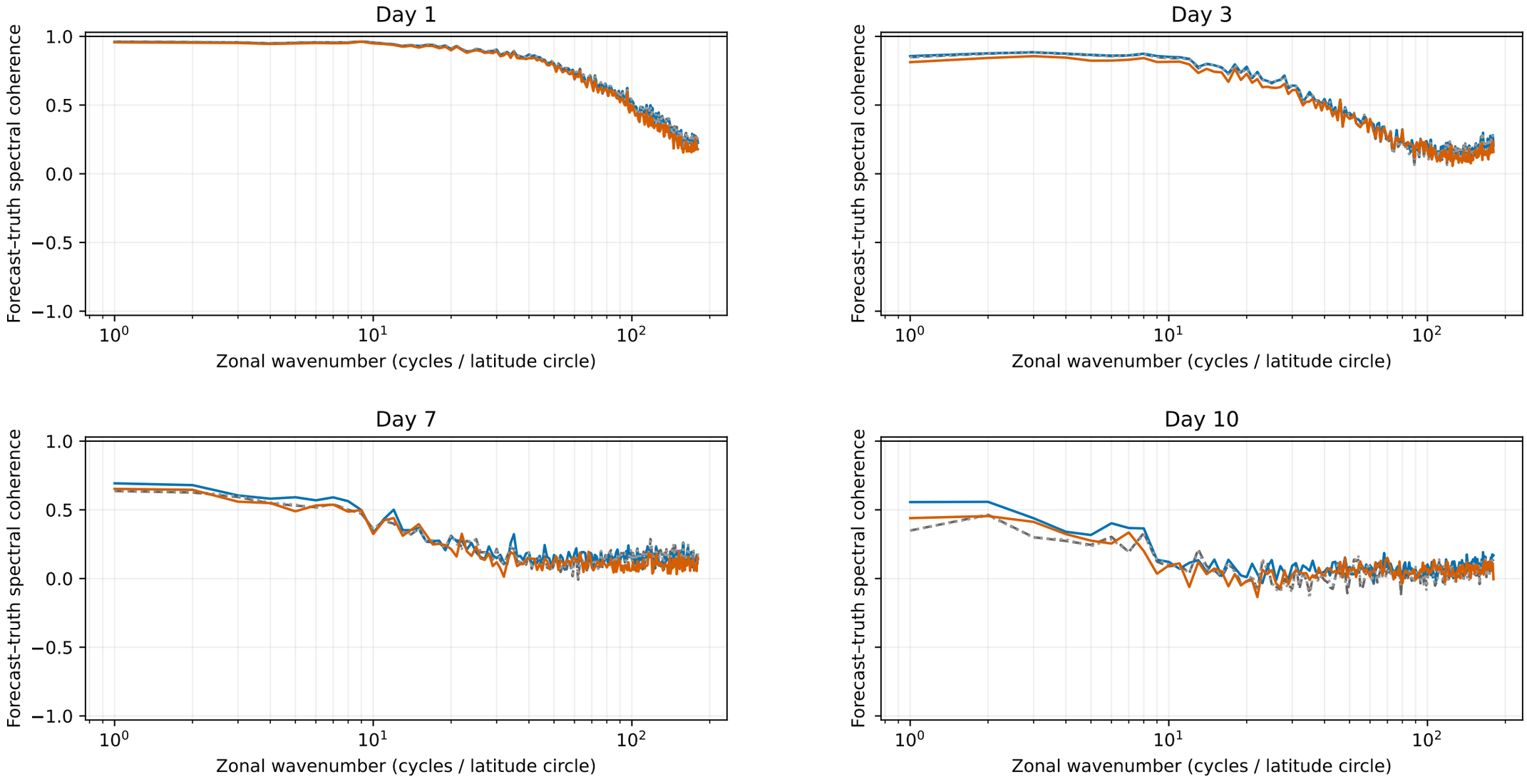

WS850: forecast–truth spectral coherence · step2000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

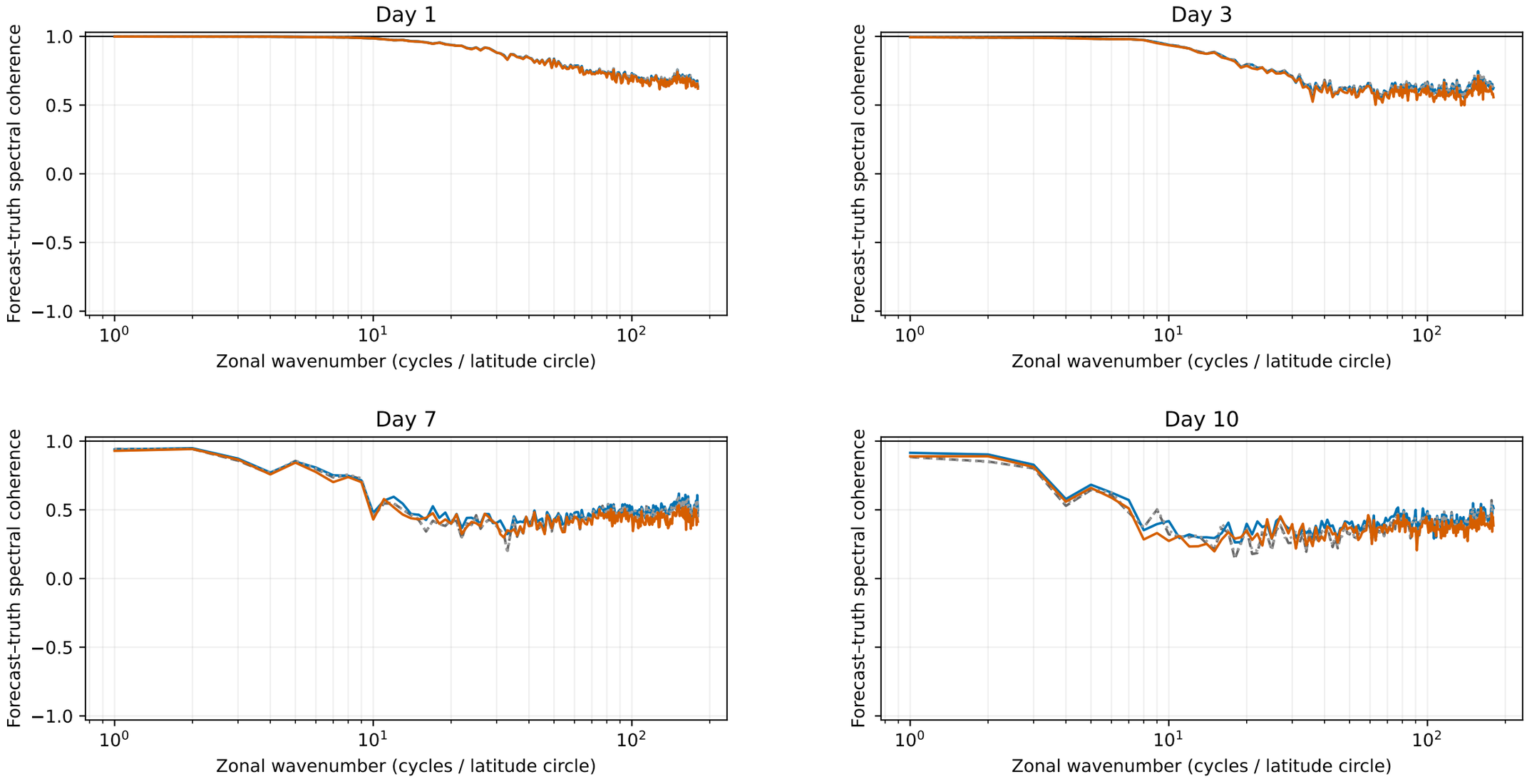

T2m: forecast–truth spectral coherence · swa_step00500-02000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

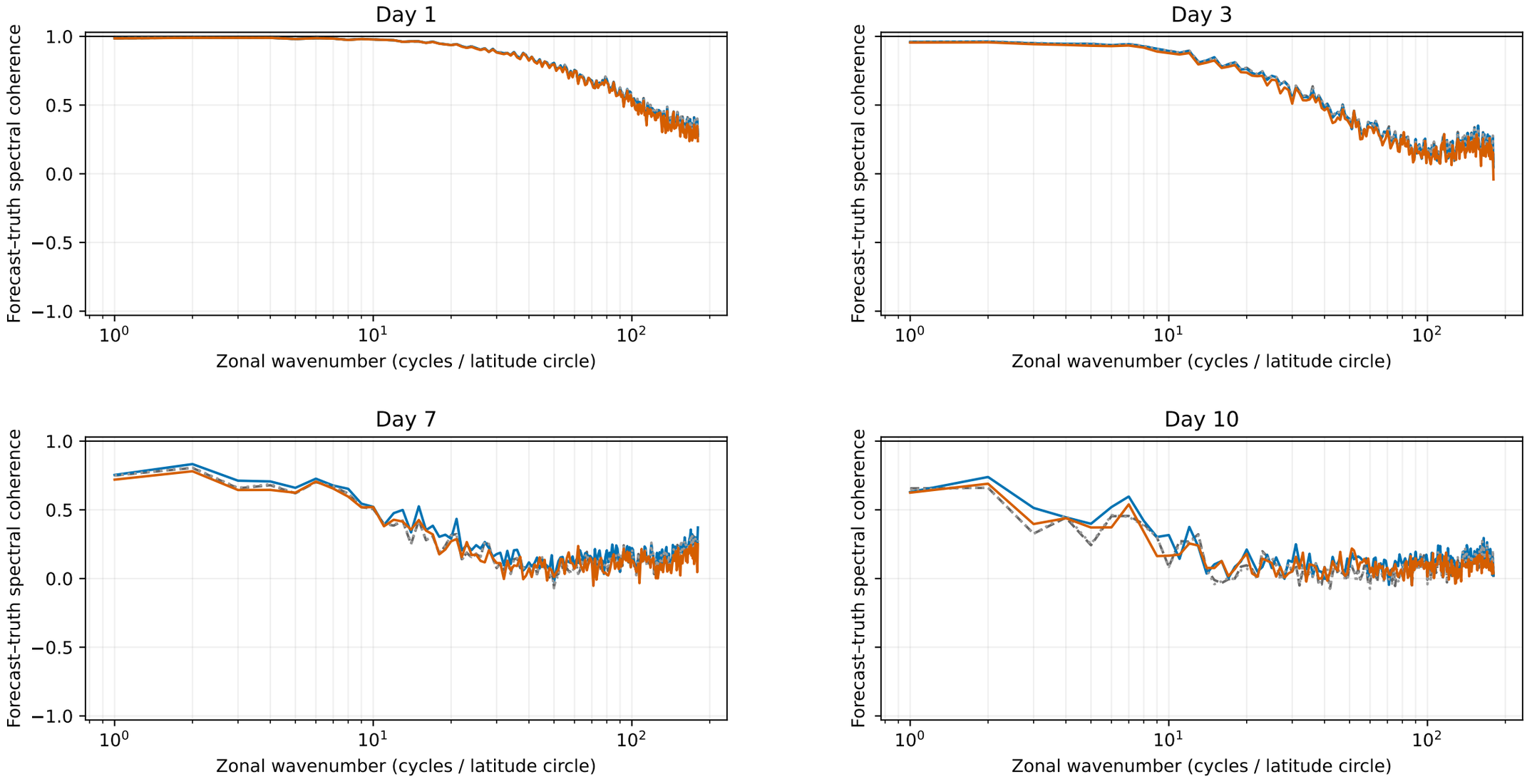

U10: forecast–truth spectral coherence · swa_step00500-02000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

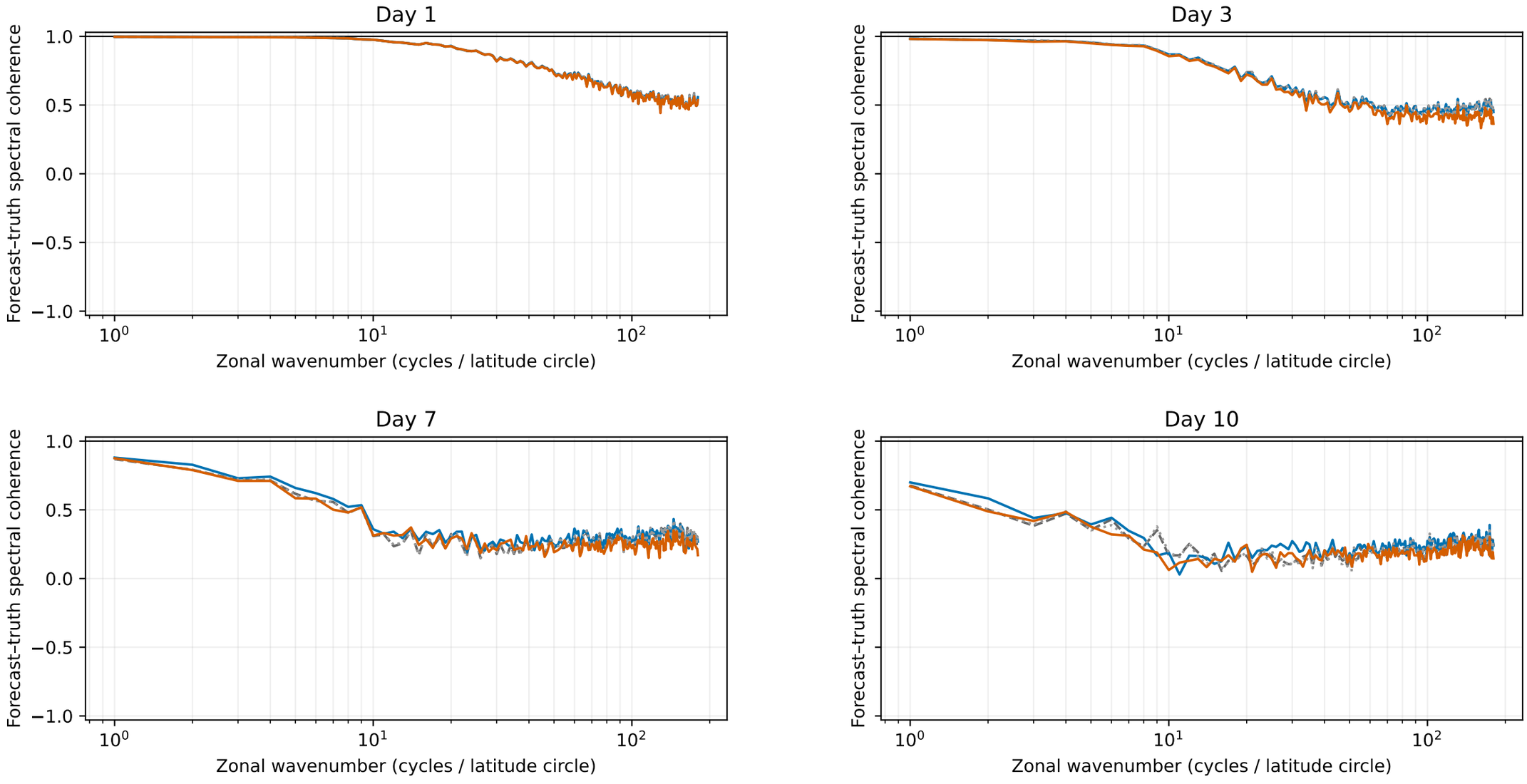

V10: forecast–truth spectral coherence · swa_step00500-02000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

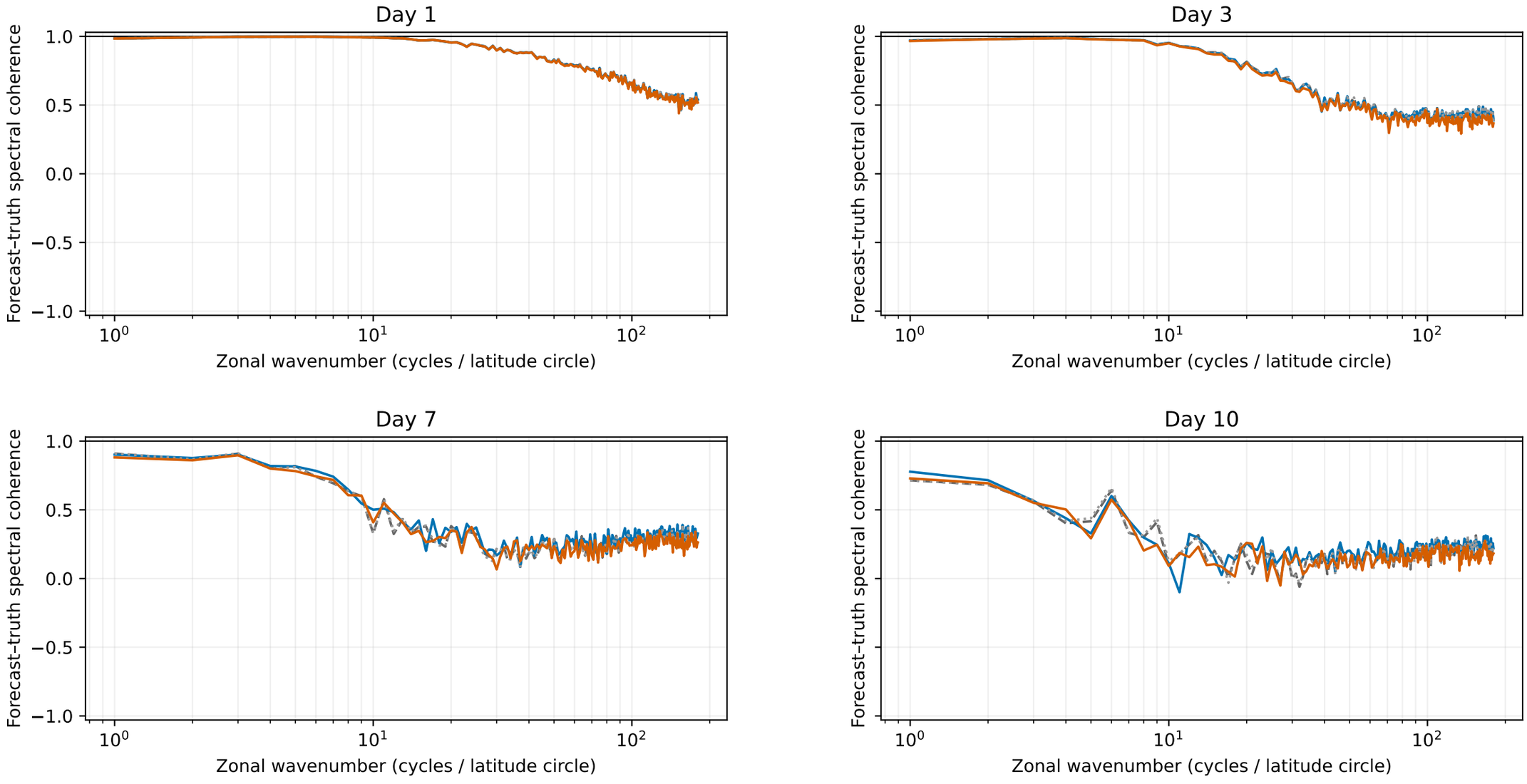

MSLP: forecast–truth spectral coherence · swa_step00500-02000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

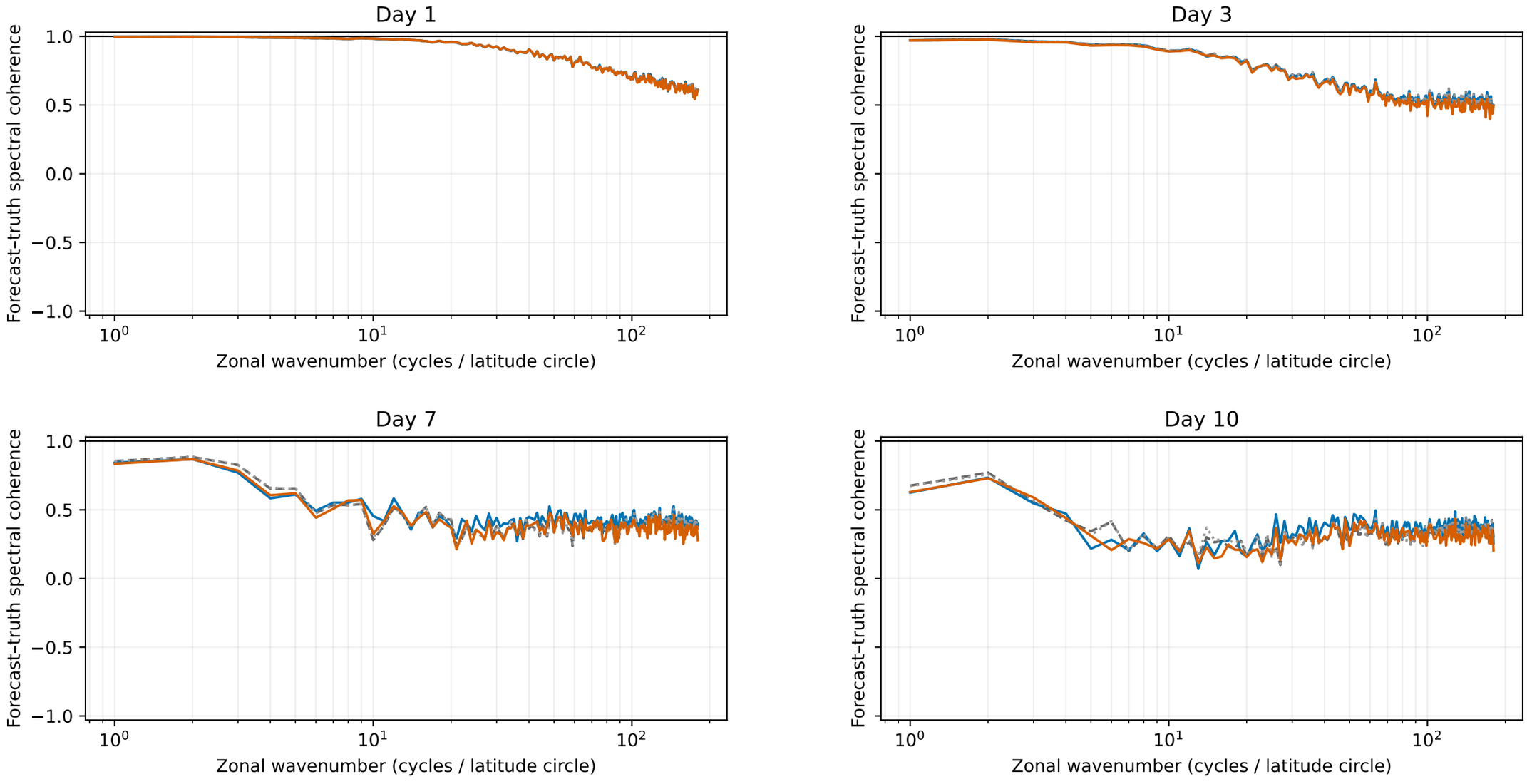

TP6: forecast–truth spectral coherence · swa_step00500-02000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

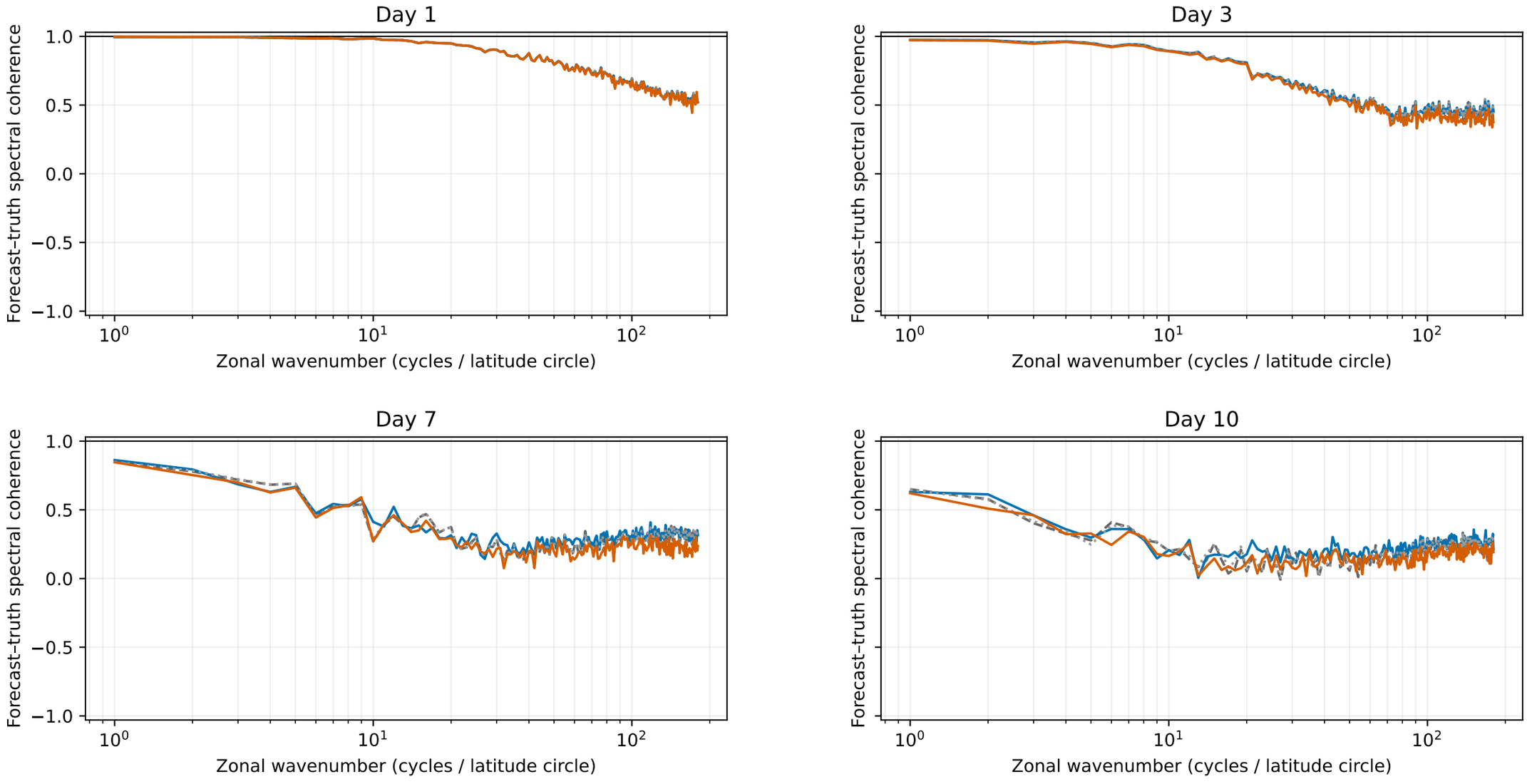

T850: forecast–truth spectral coherence · swa_step00500-02000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


Q700: forecast–truth spectral coherence · swa_step00500-02000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


U850: forecast–truth spectral coherence · swa_step00500-02000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


V850: forecast–truth spectral coherence · swa_step00500-02000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


WS10: forecast–truth spectral coherence · swa_step00500-02000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


WS850: forecast–truth spectral coherence · swa_step00500-02000
Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better
Day 1
Day 3
Day 7
Day 10
Forecast–truth spectral coherence
Zonal wavenumber (cycles / latitude circle)
1.0
0.5
0.0
−0.5
−1.0
10^0
10^1
10^2
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.